

Question ID: 05195d8a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Medium

Question

In a sample, 80% of the items are faulty. There are 88 faulty items in the sample. How many total items are in the sample?

Correct Answer: 110

Rationale

The correct answer is 110. Let x represent the total number of items in the sample. It's given that 80% of the items are faulty and that there are 88 faulty items in the sample. Therefore, 80% of x is 88. Since 80% can be rewritten as $\frac{80}{100}$, it follows that $\frac{80}{100}x = 88$. Multiplying both sides of this equation by 100 yields $80x = 8,800$. Dividing both sides of this equation by 80 yields $x = 110$. Therefore, there are 110 total items in the sample.